

1. Document Header

Process Overview

Process Name	Employee Onboarding Processing
Bot/Project ID	TBD
Process Owner	Human Resources Department
Automation Platform	UiPath Studio Web
AI Service	Flask REST API (Render) with PDF-based Risk Policy Engine
Data Source	Google Sheets (new hire tracking) + Risk Control Policy PDF
Volume	135 hires/month 1620 hires/year
Priority	RSVP Score: 4.49/5
Date Created	Pending
Status	In Review

2. Executive Summary

Business Context

The Employee Onboarding Process ensures new hires are properly entered into HR systems, provided IT setup and access before their start date, and verified for compliance. Bearcat Corporation currently uses a manual process across multiple disconnected inputs including a new hire tracking spreadsheet, a Risk Control Policy PDF, a background check portal, and email communication.

Problem Statement

- Manual, time-consuming processing: 60 minutes per employee, totaling 135 hours/month
- High error rate: 7% documentation and access provisioning errors, causing rework
- Compliance & documentation risks due to inconsistent manual verification
- Multiple disconnected inputs: new hire tracking spreadsheet, Risk Control Policy PDF, background check portal, and email communication
- Annual cost of \$166,750 labor + \$12,500 rework = \$179,250/year

Automation Objectives

The project goal is to automate the end-to-end onboarding process using UiPath Studio Web, a Flask REST API hosted on Render, and Google Sheets. The aim is to reduce manual workload, decrease errors, accelerate IT provisioning, and centralise all results in a live Google Sheets log.

RSVP Score

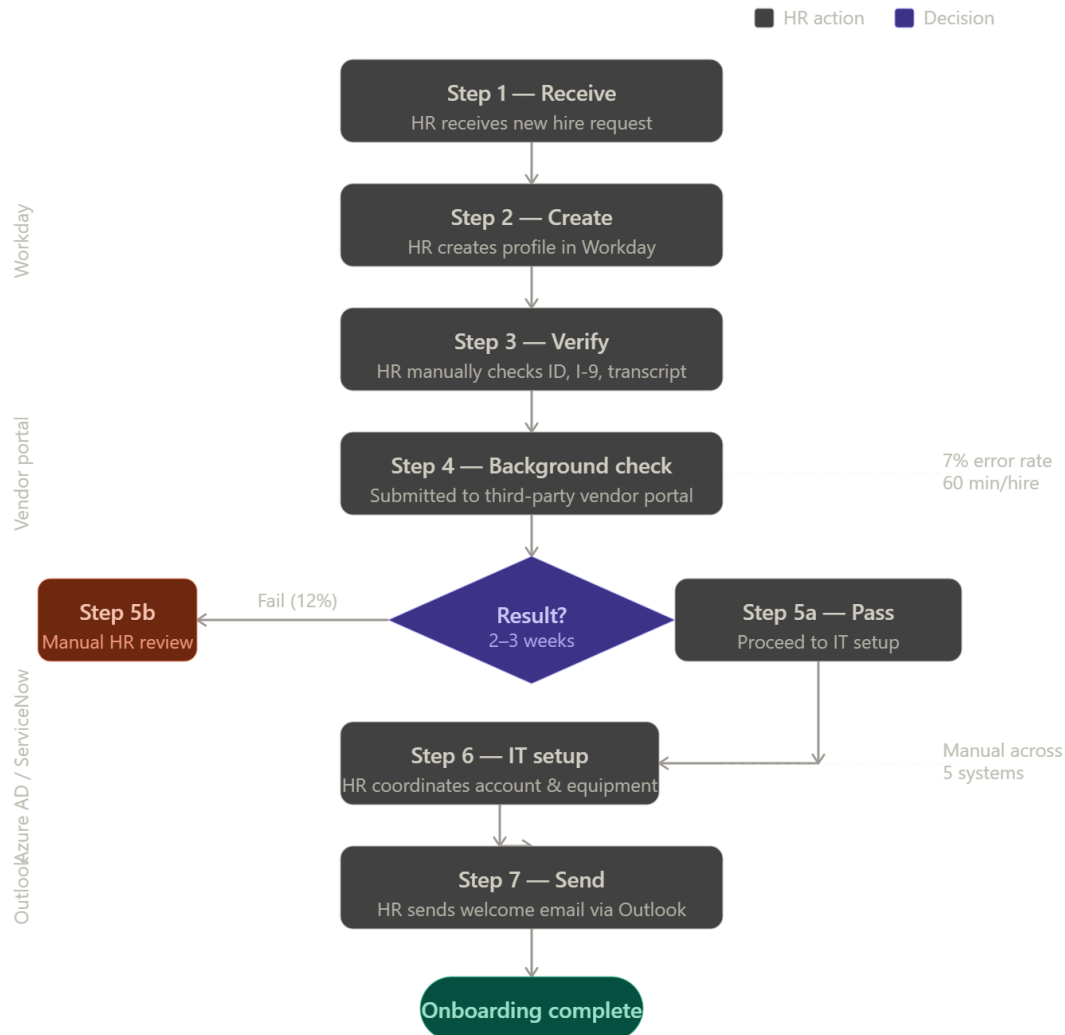
Dimension	Weight	Score	Rationale
Volume	30%	4.5/5	135 hires/month; 1,620/year, high enough to justify automation
Pain Level	25%	4.4/5	60 min/hire; 0.84 FTE consumed; 7% error rate causing rework
Stability	20%	4.3/5	Google Sheets tracking file, Flask API, and Risk Control PDF rarely change; consistent policies
Rule-Based	25%	4.7/5	Document checks, background check status, and IT setup all follow clear rules
Overall Score	N/A	4.49/5.0	High Priority - Proceed with automation

3. Current State Processes (As-Is)

- Step 1 – RECEIVE: HR receives a new hire request
- Step 2 – CREATE: HR manually creates an employee profile in the HRIS (Workday)
- Step 3 – VERIFY: HR manually verifies submitted documents (transcript, ID, agreements) via PDF
- Step 4 – BACKGROUND CHECK: The background check is submitted to a third-party vendor via a web portal. Result (Pass / Fail) is returned after 2–3 weeks.
- Step 5a – PASS: HR continues to IT setup
- Step 5b – FAIL: 12% of cases route to Manual HR Review; candidate may be contacted or rejected
- Step 6 – COMPLETE IT & Equipment Setup: HR manually coordinates IT account provisioning and equipment orders
- Step 7 – SEND: HR sends a welcome package to the new hire via the self-hosted Flask inbox (bearcat-rpa.onrender.com/inbox)

Monthly Volume	135 hires/month
Processing Time	60 minutes per employee
FTE Hours	135 hours/month (0.84 FTE)

Error Rate	7% (documentation/access issues)
Annual Labor Cost	\$166,750
Annual Rework Cost	\$12,500
Total Annual Cost	\$179,250



4. Future State Process (To-Be)

- Step 1 – BOT TRIGGER: Bot reads the New Hire Google Sheets tracking file on a scheduled basis. For each row where the background check status is not yet 'Pass', the bot waits or flags for follow-up.
- Step 2 – CROSS REFERENCE - The bot cross-references against the Risk Control PDF policy to confirm role-specific requirements are met.

- Step 3 – BACKGROUND CHECK: Bot queries the third-party vendor portal via API integration to retrieve the current status. Status is written back to the Google Sheets tracking file automatically.
- Step 4 – CHECKLIST EVALUATION: Bot evaluates all conditions: documents verified, background check passed, risk policy satisfied. If all conditions are met, the process continues automatically.
- Step 5 – IT provisioning email: Bot notified IT department for setting accounts.
- Step 6 – WELCOME EMAIL: Bot sends an automated onboarding welcome package to the new hire via Outlook.
- Step 7 – EXCEPTION HANDLING: If the background check fails, the bot sends a rejection notification. If documents are missing or invalid, the bot flags for Manual HR Review.

RPA Platform	UiPath
Data Input / Output	Google Sheets(new hire tracking file)
Policy & BG Check	Flask REST API (Render) :reads Risk Control Policy PDF; returns risk classification and BG result
IT Ticketing	ServiceNow, replicated with self-host Email Inbox,viewable at bearcat-rpa.onrender.com/inbox
Communication	Microsoft Outlook replicated with self-host Email Inbox,viewable at bearcat-rpa.onrender.com/inbox
Policy Reference	PDF Risk Control File

Key Improvements

Metric	Current	Future	Improvement
Processing Time	60 min	10 min	83% faster
FTE Hours	135 hrs/month	22.5 hrs/month	83% reduction
Error Rate	7%	1%	86% reduction
Annual Cost	\$179,250	\$52,000	71% reduction

Performance Targets

- Processing time reduced to 10–15 minutes
- Error rate reduced to below 2%
- 83% workload reduction

5. Business Rules & Decision Logic

BR-001: Background Check Routing

Rule ID	Condition	Action	Assigned To
BR-001a	Status = Pass	Proceed to IT provisioning and welcome email	Bot
BR-001b	Status = Fail	Send rejection notification; route to HR exception queue with full case context	Bot notifies; HR reviews
BR-001c	Status = Pending AND days_waiting < 15	Skip row; retry on next scheduled run (every 4 hours)	Bot
BR-001d	Status = Pending AND days_waiting >= 15	Flag for HR Manager escalation; pause onboarding workflow	Bot flags; HR Manager

BR-002: Risk Policy Compliance Rules

Risk policy classification is determined by the Flask API reading the Risk Control Policy PDF (RCP-HR-2025-001). The API returns a risk level (HIGH / MEDIUM / STANDARD) and required documents for each role. If the background check fails for any role, the bot sends a rejection email regardless of risk level.

BR-003: IT Provisioning Rules

Rule ID	Condition	Action	Notes
BR-003a	All checks passed	Bot sends IT provisioning email	Sent via self-hosted Flask inbox

		including role and equipment details	
BR-003b	Email send succeeds	Bot generates IT Ticket ID (IT-EMP###-YYYYMMDD) and writes it to Google Sheets Column U	Bot-generated; not system-assigned
BR-003c	Provisioning complete	Bot sends welcome email and updates Google Sheets with Welcome_Email_Sent = Yes	Final step for process closure

6. Exception Handling

Background Check – Fail	Vendor returns 'Fail' status	Bot sends rejection email; notifies HR
Background Check – Pending	Status not yet returned (2–3 wks)	Bot skips row; retries on next scheduled run
Document Validation Failure	Any Doc_*_Received = No	Bot sends HR notification; skips provisioning ; HR contacts candidate to re-submit missing documents
Flask API Unavailable	API returns error or timeout	Try-Catch logs error; row skipped; retried next run
Sheets Write Fails	Google Sheets API error	Try-Catch logs error; continues to next row

Out of Scope

- Employee termination, transfer, international onboarding, or offboarding
- Direct deposit configuration, benefit enrollment, payroll, and W-2 setup
- Rehire cases with pre-existing employee records and integration for training assignments

Future Considerations

- Extend the automation to include benefit enrollment notifications
- Implementation of a real-time dashboard for showing the onboarding pipeline
- Explore Gen AI for sending automated welcome messages

7. Data and System Mapping

Data Sources

Type	System / Format	Owner	Purpose
New Hire Tracking File	Google Sheets	HR Operations	Primary trigger input: includes employee info, BG check status, and write-back results.
Risk Control Policy	PDF (RCP-HR-2025-001)	Compliance Dept.	Defines compliance rules and role-based document requirements; accessed via Flask API.
Background Check	Flask REST API (Simulated)	Third-Party Vendor	Provides simulated background check statuses: <i>Pass / Fail / Pending</i> .
Employee Documents	PDF / Email Attachment	Applicant Submission	Contains necessary onboarding files: Transcript, ID, I-9, and agreements.
Email Inbox	Built-in Flask Inbox (/inbox)	Bot Output	Central repository for all bot-generated emails; viewable at bearcat-rpa.onrender.com/inbox .

System Mapping

System	Integration Method	Direction	Trigger
Google Sheets	Google Sheets Connector (OAuth)	Read + Write	Scheduled polling; write-back after each row

Flask REST API	HTTP Request Activity	Read (BG status, policy)	Per new hire row processed
Risk Control PDF	Flask API (server-side read)	Read	Embedded in Flask API; rules extracted at runtime
Built-in Email Inbox	HTTP POST to /send-email	Write (send email)	After BG Pass, Fail, or escalation trigger
Google Sheets (log)	Write Cell Activity	Write	After every row processed — columns I through AA

Data Validation Rules

- All Doc_*_Received fields must equal 'Yes' before provisioning proceeds
- Bot-generated IT Ticket ID format: IT-EMP###-YYYYMMDD
- Employee ID must be unique
- Background must equal "Pass" before provisioning
- All transactions are logged for audit purposes

8. Assumption and recommendation

Recommendation

Based on the analysis of the current state and the projected future state, it is strongly recommended to proceed with the automation of the Employee Onboarding process using UiPath Studio Web, Flask REST API, and Google Sheets . The process presents an ideal automation candidate given its high volume, rule-based nature, and high cost and error burden.

Justification

- High RSVP score of 4.49/5 confirms this process is well-suited for RPA automation
- 83% reduction in processing time (60 min to 10 min per hire) directly improves new hire experience and HR productivity
- 86% reduction in error rate (7% to 1%) reduces rework, compliance risk, and downstream delays
- The payback period of approximately 5.6 months demonstrates a compelling and rapid return on investment
- Google Sheets provides a live, auditable log of all bot processing results
- Exception handling design ensures that edge cases (failed background checks, missing documents, system outages) are managed reliably without manual intervention for standard cases

ROI Summary

Item	Value	Notes
Current Annual Cost	\$179,250	Labor + Rework
Future Annual Cost	\$51,810	Labor (reduced) + Licensing + AI Services
Annual Net Savings	\$127,440	71% cost reduction
Implementation Cost	\$60,000	Development, Testing, Training
Payback Period	~5.6 months	0.47 years

Assumptions

- UiPath Studio Web Google Workspace connector is linked to a Google account with edit access to the new hire tracking sheet
- Flask REST API is deployed on Render and the /inbox endpoint is kept open to prevent free-tier sleep timeout
- The Risk Control Policy PDF (RCP-HR-2025-001 v2.2) is encoded in the Flask API and will not change within the first 12 months of deployment
- Background check status is simulated by the Flask API for demo; production migration requires live vendor API credentials
- Google Sheets new hire tracking file uses a consistent column schema; columns I-AA are reserved for bot write-back

Out of Scope

- Employee termination, transfer, International onboarding, or offboarding
- Direct deposit configuration, benefit enrollment, payroll, and W-2 set up
- Rehire cases with pre-existing employee records and integration for training assignments

Next Steps

- Complete test case execution against the Google Sheets tracking file and document bot write-back results for all branches
- Prepare and rehearse the live bot demonstration for the final presentation
- If moving to production: replace the simulated Flask background check with a live vendor API and upgrade Render to a paid tier for uptime reliability